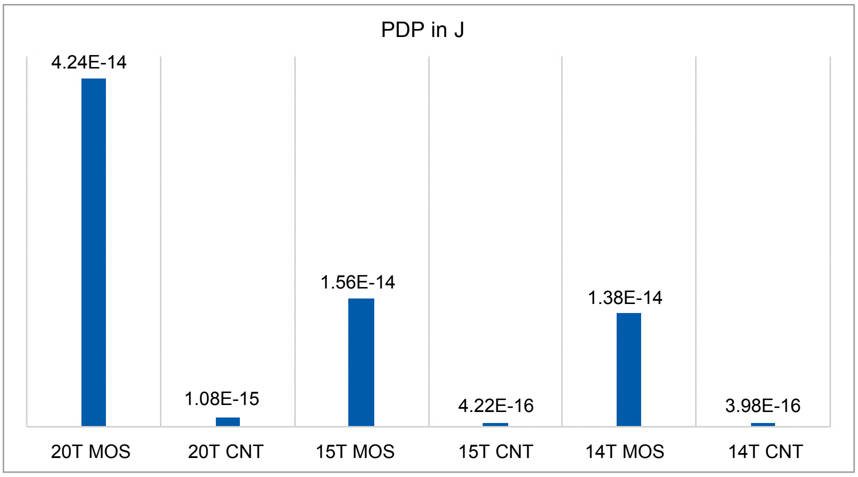

Fig. 11: PDP comparison

Simulation Results — **Table 3**

Parameters	14T MOS	14T CNT	15T MOS	15T CNT
Average Power in W	1.68E-07	4.85E-09	1.89E-07	5.15E-09
Delay in s	8.22E-08	8.21E-08	8.22E-08	8.20E-08
PDP in J	1.38E-14	3.98E-16	1.56E-14	4.22E-16
Power Dissipation in W	1.84E-07	5.77E-11	1.85E-07	7.27E-11

tion, PDP, and delay are also shown in Tables 2 and 3 where CMOS based 2 to 4 decoder is compared with the CNTFET proposed 14T and 15T decoders.

In the 20-transistor model, according to the obtained results, CNTFET based 2 to 4 decoder presents less power dissipation, less delay, and a better PDP against the 20T CMOS based decoder. Thus, overall it outperforms the conventional CMOS based decoders in all respects.

Similar results are seen for the CNTFET based mixed logic designed decoders (14T and 15T) when compared with the CMOS based mixed design decoders. Although all CNTFET models have less power dissipation and PDP than the conventional CMOS, the CNTFET based 14T performs extremely well in comparison to all other models with least power dissipation, average power consumption, and PDP.

Conclusion

Carbon nanotube field effect transistor (CNTFET) is the best amongst upcoming technologies. However, as the quality of a CNTFET is not the same as of traditional mass-produced CMOS, there is need for new production techniques.

The proposed decoders were successfully simulated in HSPICE. The results obtained confirmed proper functioning of the decoder. It was found that the proposed CNTFET circuits consume much less power, reduce transistor count, and dissipate less power as compared to MOSFET based decoders.

Simulation results show that CNTFET 2 to 4 decoder in 20T, 14T, and 15T is better than MOSFET based decoders as average power requirement is reduced substantially. Also, there is significant improvement in the PDP and power dissipation voltage source when compared with an MOSFET decoder.

Therefore, we finally conclude that the mixed logic CNTFET decoder is very much effective for reducing the transistor count, average power, and power dissipation. EFY

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